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Study on Fracture Propagation and Parameter Optimization for Multi-Cluster Fracturing Using Temporary Plugging Technology——A Case Study in the Tight Oil Reservoir, Sichuan Basin

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Abstract

Multi-cluster staged fracturing is an important measure to exploit unconventional resources. During fracturing process, the fracturing fluid mainly flows into few clusters, resulting in uneven fracture propagation. The temporary plugging technology is generally applied to balance the fluid distribution into each cluster within a stage. However, plugging timing, frequency, and amount of plugging balls have not been clearly revealed. In this paper, a model of three dimensional (3D) fracturing model is established, and this model is verified by the indoor fracturing experiment and traditional analytical solution. The multiple fracture propagation under different perforation parameters and temporary plugging parameters is studied, and the variation coefficient of fracture length is applied to quantitatively evaluate the fracture uniformity. The simulation results show that the inner fractures are more easily suppressed by outer fractures due to stress interference. When the plugging balls are used, the outer fractures will be preferentially blocked, and

the fracturing fluid into these clusters begins to decrease. In this case, the subsequent fluid will flow into the inner fractures to enhance their fracture length, finally achieving even stimulation. When the number or diameter of perforation holes is lower, the more uniform fracture geometry can be obtained due to limited-entry effect, and the optimal perforation number is 6 per cluster, and the diameter of perforation hole is recommended to not exceed 8mm. When there are 5 clusters within a stage, only one plugging operation is needed, and for 6 or 7 clusters, two plugging operations are necessary to ensure the uniform fracture propagation. With the increase of number of plugging balls, the variation coefficient of fracture length first decreases and then increases, and the optimal amount of plugging balls is 16-20. It is suggested to pump the plugging balls when 60-70% of fracturing fluid is injected. This method proposed in this paper is applied in Well X1 in the Sichuan Basin, China, and the variation coefficient of fracture length is controlled less than 10%, obtaining good application results.

Introduction

Following the success of shale gas development, tight oil has become a focal point in global unconventional oil and gas exploration. In China, tight oil reservoirs account for over 75% of newly added proven oil reserves, demonstrating substantial potential for future energy supply (Zhu et al., 2019). Tight oil reservoirs exhibit poor physical properties with low matrix permeability, making conventional fracturing techniques inadequate for commercial development. To address these challenges, the multi-cluster staged fracturing technology has emerged as a dominant approach. This method creates multiple complex fractures from the wellbore, significantly reducing flow resistance and enhancing oil production (Xu et al., 2018). However, tight oil and gas reservoirs have strong heterogeneity in physical properties and stress. Under multi-cluster perforation conditions within a stage, the fluid flowing into each cluster varies greatly under the combined effect of reservoir heterogeneity and stress interference, resulting in uneven fracture initiation and propagation, which seriously affects the effectiveness of reservoir stimulation (He et al., 2023). Field monitoring data from distributed acoustic sensing (DAS) and distributed temperature sensing (DTS) also reveal significant uneven fluid distribution during fracturing operations. Approximately 33% of stimulated clusters have almost no production contribution (Miller et al., 2011; Somanchi et al., 2017). Recent field trials conducted in China's tight oil reservoirs, including downhole video and DAS monitoring, had yielded consistent observations (Zhang et al., 2025). As a kind of tool-free method, temporary plugging technology has emerged as an effective solution to improve fracture uniformity in tight oil horizontal wells. The technique involves pumping spherical or knot-type diverters to temporarily block preferred-entry perforation clusters, thereby increasing wellbore net pressure and redirecting subsequent fluid to under-stimulated or unstimulated clusters.

To better understand the influence of temporary plugging on hydraulic fracture propagation, researchers have conducted laboratory-scale true triaxial fracturing experiments. Among them, Shi et al. (2020) investigated fracture propagation in high-permeability unconsolidated sandstone using oil-soluble diverters and monitored fracture propagation via acoustic emission (AE) technology. The results indicate that there are three typical fracture shapes after fracturing: multi-fracture, single-wing fracture, and by-wing fracture. Zhang et al. (2020) conducted true triaxial fracturing experiments using processed five types of shale outcrops. Their study specifically examined how horizontal stress anisotropy and diverter concentration influence both plugging efficiency and fracture propagation patterns. Xu et al. (2024) conducted a series of plugging experiments to investigate the plugging performance of different diverter materials including fiber, particle, and fiber-particle combination materials. However, current laboratory studies primarily focus on investigating the influence of temporary plugging on the propagation behavior of a single fracture. Due to factors such as experimental scale effects, these studies fail to adequately reveal the propagation mechanisms of multi-cluster fracture propagation in horizontal wells under temporary plugging conditions. Numerical simulation has become a crucial approach for investigating the competitive propagation of multiple fractures in horizontal wells. Employing particle-based discrete element method (DEM), Wang et al. (2025) revealed the mechanism of fracture propagation through analyzing the induced stress, and investigated the effect of geo-stress and fracturing fluid viscosity on the fracture geometry. Li et al. (2025) studied the effects of natural fractures, cluster space, and pumping rate on the propagation of multiple fractures using finite element method (FEM). Tao et al. (2024) investigated the influence of critical parameters, including horizontal stress differences, fracturing fluid viscosity, injection rate, and initial fracture angle, on fracture propagation using extended finite element method (XFEM) to analyze multi-stage temporary plugging fracturing in tight oil reservoirs.

However, previous researches have predominantly focused on mechanistic investigations, which fail to provide effective guidance for field operations. The optimization of temporary plugging parameters still largely relies on empirical approaches, lacking a quantitative design methodology. To address this problem, this study develops a three-dimensional (3D) multiple fracture propagation model incorporating temporary plugging effects based on the 3D discrete lattice method. This model can consider fluid-mechanical coupling and stress interference effect. The proposed model systematically investigates fracture propagation characteristics before and after temporary plugging under various parameters. Perforation parameters and temporary plugging parameters are quantitatively optimized for horizontal well fracturing. The model has been validated through successful application in designing the fracturing operation for a tight oil well in the Sichuan Basin. The research results allows us to gain a better insight on multiple fracture propagation considering temporary plugging.

3D discrete lattice method

In this paper, a 3D discrete lattice method based on synthetic rock mass technology and discrete lattice theory is applied to simulate the multiple fracture propagation considering the temporary plugging process (Damjanac et al., 2016). The methodology employs a bonded particle model to discretize rock masses into nodes, where spring elements represent elastic interactions at rock contact interfaces. The smooth joint model is implemented to simulate the pre-existing discontinuous weak planes within the rock matrix. The lattice unit is connected through springs, and a pipe network can be formed between the fluid units located at the center of the spring to describe fluid flow. A lattice spring network is composed of numerous quasi random distributed nodes connected by springs, and joints can be placed in any orientation to accurately and efficiently characterize cracks (Huang et al., 2020).

The model employs an explicit dynamic algorithm to directly simulate joint behaviors including fracture initiation, opening, closure, and slippage, demonstrating superior numerical stability. Each node incorporates three translational and three rotational degrees of freedom, with displacement evolution governed by the central difference equation:

$$\begin{cases} \dot{u}_i^{(t+\Delta t/2)} = \dot{u}_i^{(t-\Delta t/2)} + \sum F_i^{(t)} \Delta t / m \\ u_i^{(t+\Delta t)} = u_i^{(t)} + \dot{u}_i^{(t+\Delta t/2)} \Delta t \end{cases} \tag{1}$$

where $\dot{u}_i^{(t)}$ is the velocity of component i (i=1,2,3) at time t, m/s, $u_i(t)$ is the displacement of component i at time t, m, $F_i^{(t)}$ is the force of component i acting on the node at time t, N, Δt is the time step, s, and m is the mass of the node, kg.

The angular velocity of component i at time t is calculated by the following equation:

$$\omega_i^{(t+\Delta t/2)} = \omega_i^{(t-\Delta t/2)} + \frac{\sum M_i^{(t)}}{J} \Delta t \tag{2}$$

where $\omega_i^{(t)}$ is the angular velocity of component i at time t, rad/s, $M_i^{(t)}$ is the moment of component i at time t, N·m, and J is the moment of inertia, kg·m^2.

The relative displacement of nodes is applied to calculate the force variation acting at springs:

$$\begin{cases} F_{N,i} \leftarrow F_{N,i} + \dot{u}_{N,i} k_{N,i} \Delta t \\ F_{s,i} \leftarrow F_{s,i} + \dot{u}_{s,i} k_{s,i} \Delta t \end{cases} \tag{3}$$

where $F_{N,i}$ and $F_{s,i}$ are the normal and tangential stress of component i acting at the node, respectively, N, $\dot{u}_{N,i}$ and $\dot{u}_{s,i}$ are the normal and tangential node velocity of component i, respectively, m/s, $k_{N,i}$ and $k_{s,i}$ are the normal and tangential stiffness of component i, respectively, N/m.

The correspondence between the tensile and shear strength of micro springs and macro rock masses can be described as:

$$\begin{cases} F_{N,\max} = a_t T R^2 \\ F_{S,\max} = \mu F_{N,\max} + a_s C R^2 \end{cases} \tag{4}$$

where $F_{N,\max}$ and $F_{S,\max}$ are the tensile and shear force of the spring, respectively, N, a_t and a_s are the correction coefficients of tensile and shear strength, respectively, T and C are the tensile and shear strength of macroscopic rock mass, respectively, Pa, R is the lattice cell size, m, and μ is the friction coefficient.

When the normal stress of a spring exceeds its tensile strength or the shear stress exceeds its shear strength, the tensile or shear failure of spring occurs. After the spring is damaged, micro-cracks are generated, and the force acting on the spring is zero, which can be described as $F_N = F_S = 0$.

The fluid flow within cracks is numerically solved using a pipe network model of interconnected fluid units, where the fundamental flow between adjacent elements A and B follows the modified cubic law formulation (Batchlor et al., 1967):

$$q = \beta k_r \frac{w^3}{12\mu}\left[p_A - p_B + \rho_w g (z_A - z_B)\right] \tag{5}$$

where q is flow rate, m^3/s, β is the dimensionless correction factor, k_r is the relative permeability, w is the fracture width, m, p_A 、p_B are the fluid pressure at node A and node B, respectively, Pa, ρ_w is the fluid density, kg/m^3, and g is the gravitational acceleration, m/s^2, and z_A 、z_B are the height of node A and node B, respectively, m.

The flow evolution model is solved using explicit solution method. Within the flow time step, the formula for calculating the increase in flow pressure is:

$$\Delta p = \frac{\sum_i q_i}{V} K_f \Delta t_f \tag{6}$$

where Δp is the increment of fluid pressure, Pa, K_f is the bulk modulus of fluid, GPa, V is the fluid volume at nodes, m^3, and q_i is the flow rate of the i-th node, m^3/s.

The fluid flow distribution into individual hydraulic fracture exhibits dynamic variations due to stress interference between adjacent fractures and pressure drops across perforation holes. Under the constraint of mass conservation, the sum of flow rates through all clusters must equal the total injection rate. This relationship is mathematically expressed as:

$$q_T = \sum_{i=1}^{N} q_i \tag{7}$$

where q_T is the total injection rate, m^3/s, N is the cluster number, and q_i is the injection rate flowing into the i-th perforation cluster, m^3/s.

The pressure drop loss is described using Bernoulli equation (Tang et al., 2023):

$$P_{pf} = 0.807249 \frac{\rho}{n^2 D^4 C_d^2} q_i^2 \tag{8}$$

where P_{pf} is the pressure drop loss of the i-th perforation cluster, Pa, ρ is the fracturing fluid density, kg/m^3, n is the number of perforation holes in the i-th perforation cluster, D is the diameter of perforation hole, m, and C_d is the flow coefficient of perforation hole.

In order to make the temporary plugging process closer to engineering reality, during the calculation process, it is necessary to gradually reduce the injection rate of the preferred-entry perforation clusters, so as to gradually restore the process of effective bridge plugging formed by the temporary plugging material near the perforation holes until the fluid flowing into these perforation clusters drops to zero. Based on this, the temporary plugging process for multi-cluster fracturing is simulated.

Model validation

The accuracy of the 3D discrete lattice-based fracturing model was evaluated against Dontsov's analytical solution of 3D Penny-shaped fracture (Dontsov et al., 2017). As illustrated in Fig. 1, the calculated fracture widths from both methods show excellent consistency, and the maximum error is only 0.232mm.

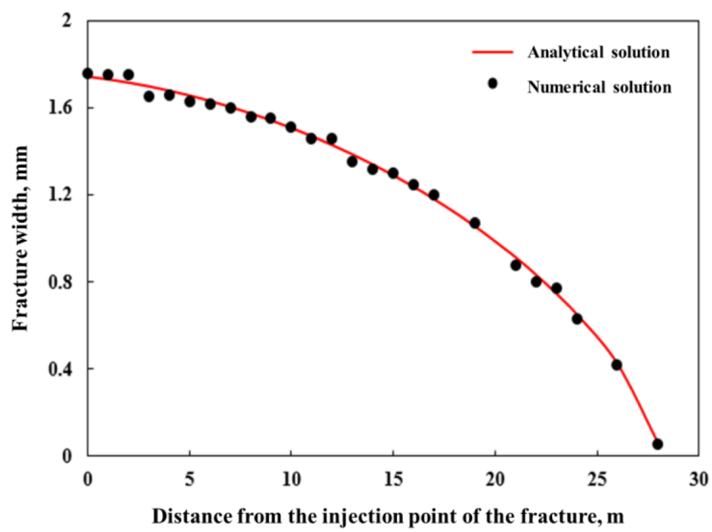

Figure 1—Model verification using the analytical solution of penny-shaped fracture propagation model (He et al., 2023)

Using the 3D discrete lattice method, Fu et al. (2022) established a tight sandstone numerical model at the same scale as the laboratory true triaxial experiment. The input rock mechanical parameters and fracturing parameters in the numerical simulation were kept consistent with those in the experimental tests. The results (see Fig. 2) demonstrate that the error between the simulated pressure curve and the experimental curve is less than 10%, and the fracture morphology closely matches the final experimental fracture pattern. The hydraulic fracture propagates along the direction of the maximum horizontal principal stress, exhibiting a single, vertically radial fracture morphology, which corresponds to the classical penny-shaped model. It can be seen that the theoretical and experimental results both indicate that the 3D discrete lattice method for modeling the fracture propagation is reliable.

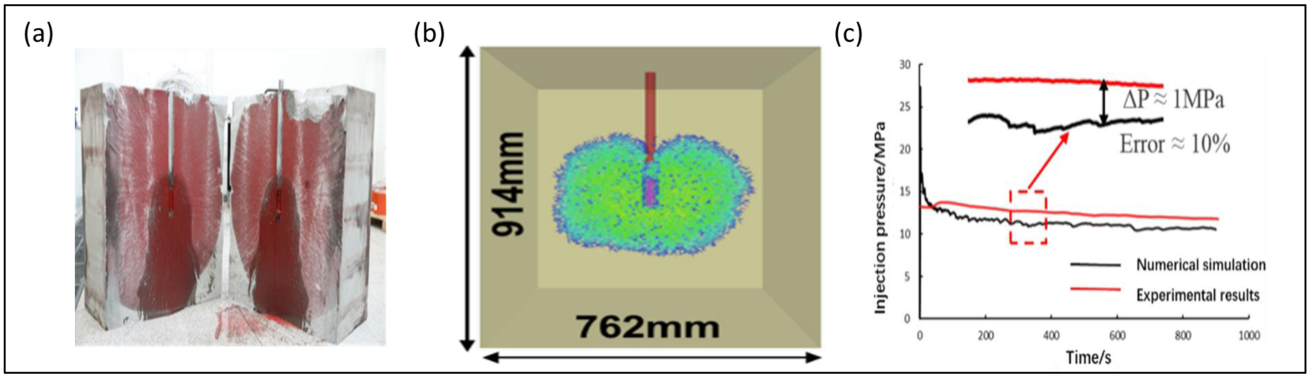

Figure 2—Model verification by true triaxial experiment: (a) Experimental fracture; (b) Numerical fracture; (c) Injection pressure curves obtained by both experiments and numerical simulation (Fu et al., 2022)

Modeling results and discussion

Input parameters

Based on the fracturing model (280m×70m) established in this study, we investigated the propagation patterns of multiple fractures under different perforation parameters (number of perforation holes, hole diameter) and temporary plugging parameters (plugging frequency, amount of plugging balls, and plugging timing). The input parameters for numerical simulations are listed in Table 1. For this research, the basical input parameters were primarily derived from a typical tight oil reservoir in the central Sichuan Basin. In each simulation group, the fundamental input parameters remained constant. Specifically, the Young's modulus is set at 35 GPa, Poisson's ratio at 0.24, and the three principal in-situ stresses ($S_H/S_v/S_h$) are 53MPa, 47MPa, and 44MPa, respectively. The fracture toughness is 0.9 MPa·$m^{0.5}$, the fracturing injection rate is 18 m^3/min, the fracturing fluid viscosity is 10 mPa·s, and the stage length is 70 m. Through comparative analysis, the optimal perforation and temporary plugging parameters were determined.

Table 1—The input parameters for simulation of multiple fracture propagation using temporary plugging

Parameters	Value	Unit
Young's modulus	35	GPa
Poisson's ratio	0.24	/
In-situ stress ($S_H/S_v/S_h$)	53/47/44	MPa
Fracture toughness	0.9	MPa·$m^{0.5}$
Fracturing injection rate	18	m^3/min
Fracturing fluid viscosity	10	mPa·s
Stage length	70	m
Number of perforation hole per cluster	2/4//6/8/10	/
Diameter of perforation hole	8/10/12	mm
Cluster number per stage	5/6/7	/
Number of plugging balls	0/8/12/16/20/24	/
Time to pump plugging balls	30%/40%/50%/60%/70%	/

(In Table 1, S_H represents the maximum horizontal principal stress, S_h represents the minimum horizontal principal stress, and S_v represents the vertical principal stress)

Evaluation objective

Fracture length is a critical parameter influencing fracturing effectiveness and a key focus in fracturing design. For multi-cluster fracturing in horizontal wells, engineers primarily aim to achieve uniform fracture lengths among clusters within each stage to ensure adequate reservoir stimulation. In this study, the variation coefficient of fracture length is employed to quantitatively evaluate the uniformity of multiple fracture propagation within a stage (see Equation 9). A lower variation coefficient of fracture length indicates more balanced fracture growth and better fracturing performance.

$$\eta = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N}(L_i - \overline{L_i})^2}}{\overline{L_i}} \times 100\% \tag{9}$$

Where η is the variation coefficient of fracture length, N is the cluster number within a stage, L_i is the fracture length of the i-th perforation cluster, m, and $\overline{L_i}$ is the average fracture length within a stage, m.

Effect of perforation parameters

Perforation number. To investigate the influence of perforation number on multiple fracture propagation, we maintain all other parameters constant in this study. Fig. 3 and Fig. 4 show the fracture morphologies and calculated variation coefficient of fracture length for five clusters per stage, respectively. The modeling results show that the reduction in perforation number leads to improved fracture uniformity. The observed results align with limited-entry perforation theory. Fewer perforations increase perforation hole friction, elevating flow resistance through the holes. This reduced pressure differential between exterior and interior fractures promotes more uniform fracture propagation within each stage. The numerical results demonstrate a clear correlation between perforation number and fracture uniformity. When the number of perforations per cluster is reduced from 10 to 6, the coefficient of fracture length decreased significantly from 40% to 15%, indicating a transition from highly heterogeneous to relatively uniform fracture geometries. However, further reduction from 6 to 2 perforations per cluster only marginally decreased the variation coefficient of fracture length from 15% to 10%, suggesting perforation number has little effect on the multiple fracture propagation when the perforation number is smaller. However, excessive reduction in perforation number would induce prohibitively higher perforation friction during fracturing operations, consequently increasing wellhead pressure. Therefore, perforation numbers must be maintained within an optimal range. In this paper, the number of perforation hole per cluster is recommended to be 6.

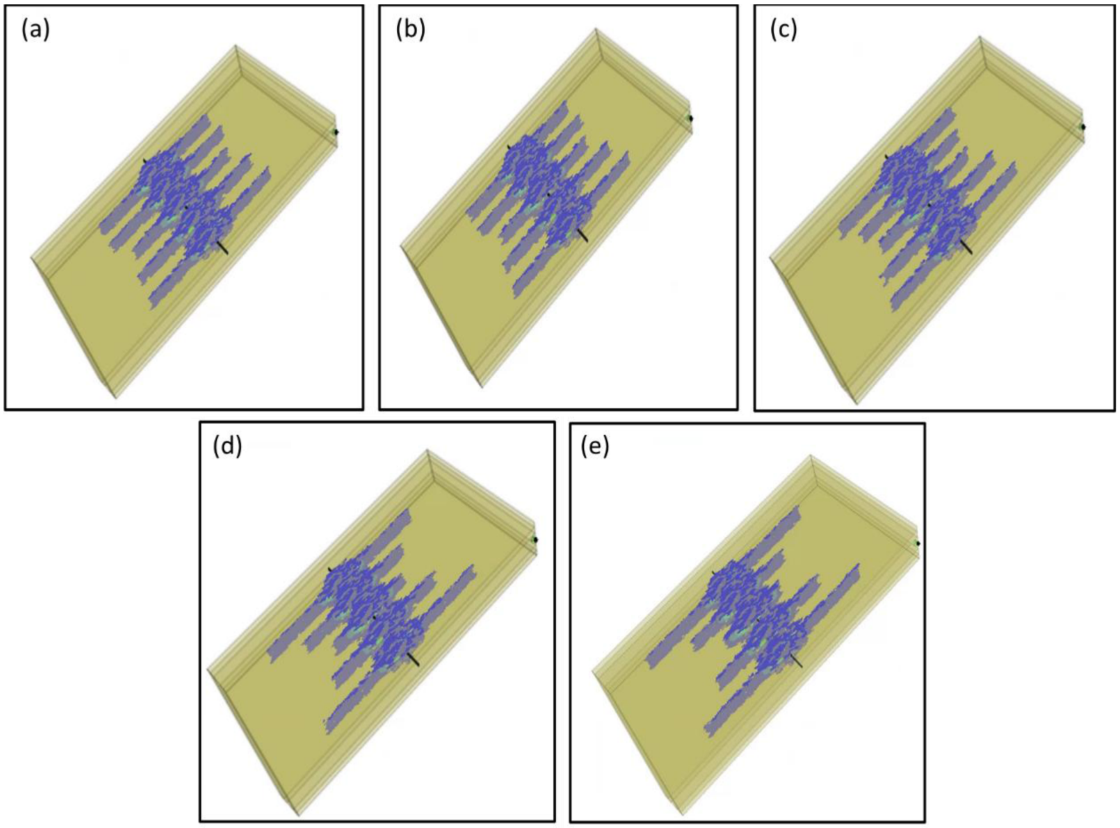

Figure 3—Fracture morphology for five cases with different number of perforation hole per cluster: (a) 2; (b) 4; (c) 6; (d) 8; (e) 10.

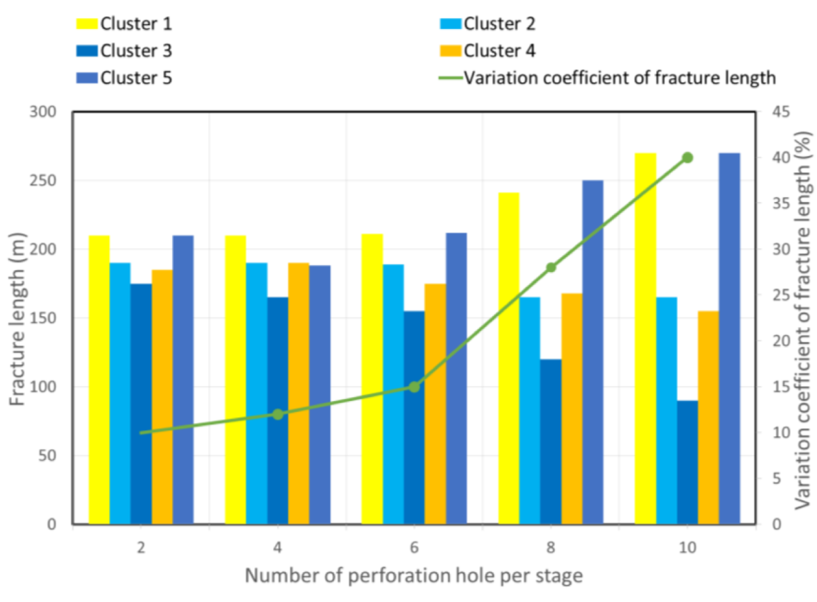

Figure 4—The variation coefficient of fracture length for five cases with different number of perforation hole per cluster: (a) 2; (b) 4; (c) 6; (d) 8; (e) 10.

Diameter of perforation hole. Fig. 5 and Fig. 6 show the fracture morphologies and calculated variation coefficient of fracture length under the condition of different diameter of perforation hole, respectively. The results indicate that a perforation diameter of 8 mm yields relatively uniform fracture geometry, with a variation coefficient of fracture length at 18%. In contrast, diameters of 10 mm and 12 mm lead to significantly increased values of 30% and 45%, respectively, demonstrating pronounced non-uniform fracture growth. This phenomenon occurs because smaller diameters generate higher localized pressure drop at the perforations, thereby promoting more balanced fluid distribution among clusters. The simulation results demonstrate that the perforation diameter should theoretically not exceed 8 mm for better fracture uniformity. However, considering that a lower perforation hole diameter can result in higher wellhead pressure, and larger proppant (30/50 mesh) is generally employed during fracturing, it is recommended to increase the hole size appropriately to ensure that the proppants do not block the perforation holes.

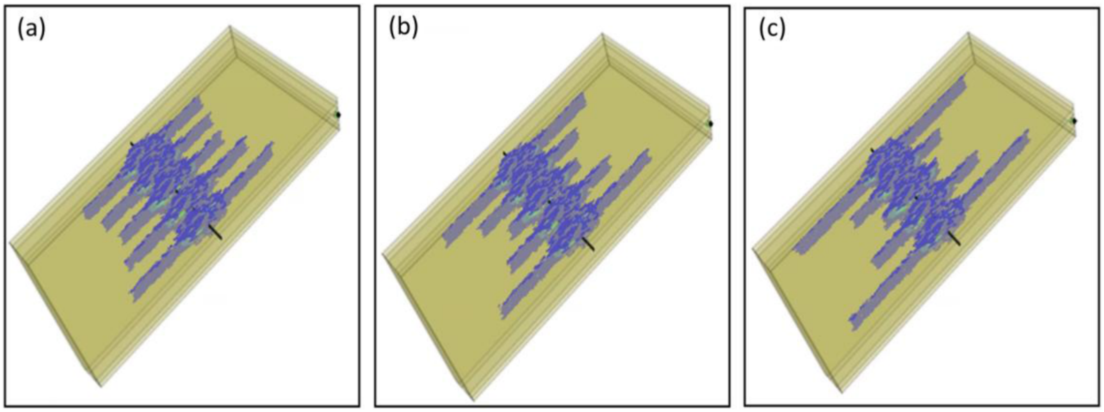

Figure 5—Fracture morphology for three cases with different diameter of perforation hole: (a) 8mm; (b) 10mm; (c) 12mm.

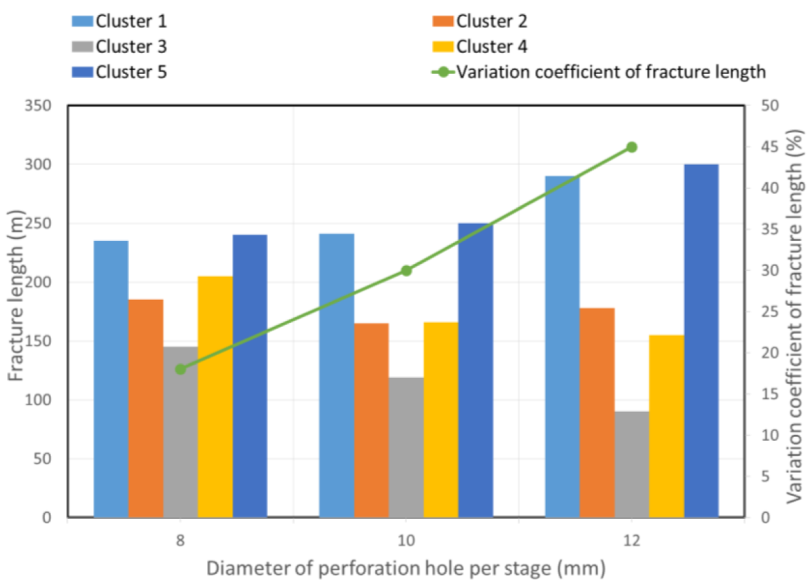

Figure 6—The variation coefficient of fracture length for three cases with different diameter of perforation hole: (a) 8mm; (b) 10mm; (c) 12mm.

Effect of temporary plugging parameters

Plugging frequency. Fig. 7 and Fig. 8 show the fracture morphologies and calculated variation coefficient of fracture length for 5 clusters under the condition of different plugging times, respectively. The results show that when only plugging once during fracturing, the variation coefficient of fracture length is only 10%, showing relatively uniform fracture propagation. As the number of plugging increases, the variation coefficient of fracture length obviously increases. When plugging twice and three times, the variation coefficient of fracture length is 15% and 25%, respectively. This inverse correlation indicates that excessive times of plugging not only prolongs operational time but also significantly reduces fracture uniformity.

Therefore, for 5 clusters within a stage, it is recommended to plug once to maintain both operational efficiency and stimulation effectiveness.

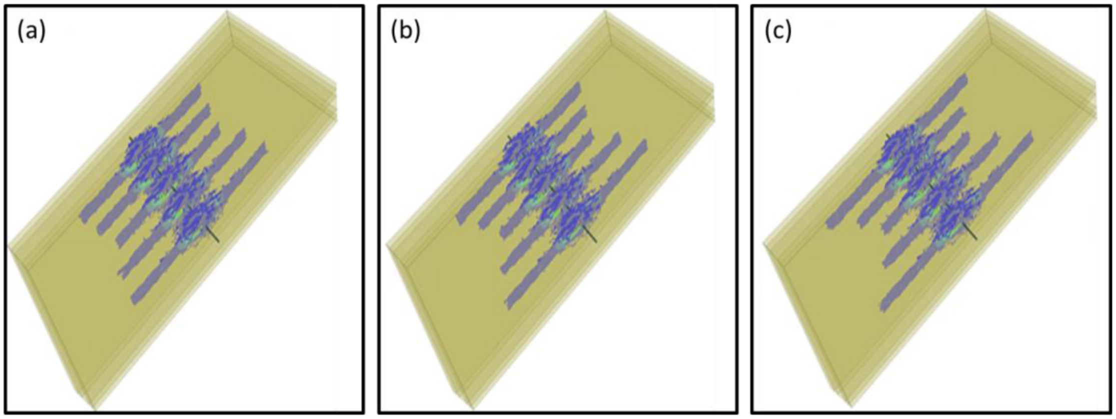

Figure 7—Fracture morphology for 5 clusters with different plugging times: (a) Plugging once; (b) Plugging twice; (c) Plugging three times.

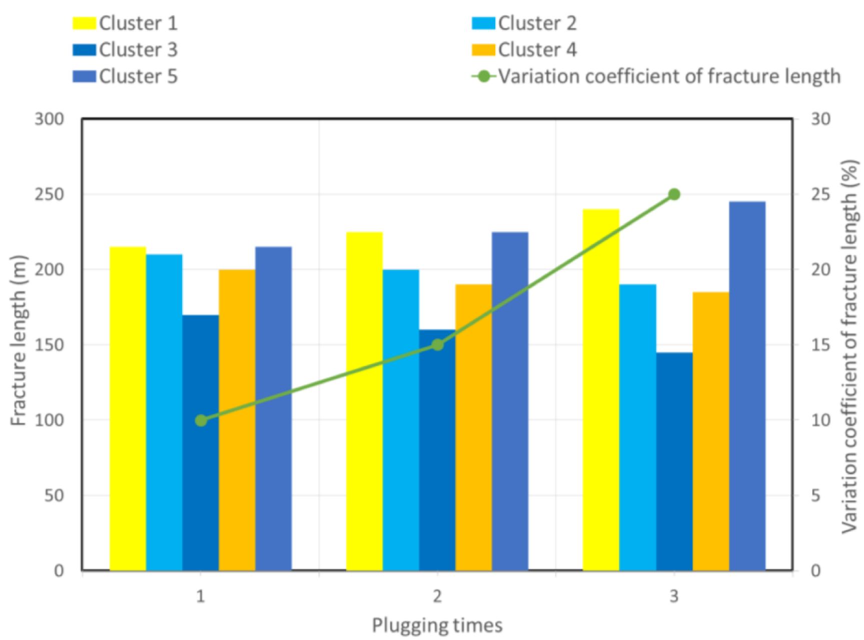

Figure 8—The variation coefficient of fracture length for 5 clusters with different plugging times: (a) Plugging once; (b) Plugging twice; (c) Plugging three times.

Fig. 9 and Fig. 10 show the fracture morphologies and calculated variation coefficient of fracture length for 6 clusters under the condition of different plugging times, respectively. The results show that when only plugging once, the variation coefficient of fracture length was high, reaching 21%, and the multiple fracture propagation is relatively uneven. Compared to the case of 5 clusters within a stage, the cluster spacing is smaller under the condition of constant stage length in this case, the stress interference effect between adjacent fractures is more obvious, and the length of the inner fractures is obviously lower than that of the outer fractures. That means plugging once can only improve the fracture uniformity to a certain extent compared with not plugging. When plugging twice, the multiple fracture propagation is significantly improved, and the variation coefficient of fracture length is reduced to 12%. Conversely, if plugging three times, the variation coefficient of fracture length will increase to 14%. Therefore, for the 6 clusters within a stage, it is recommended to plug twice.

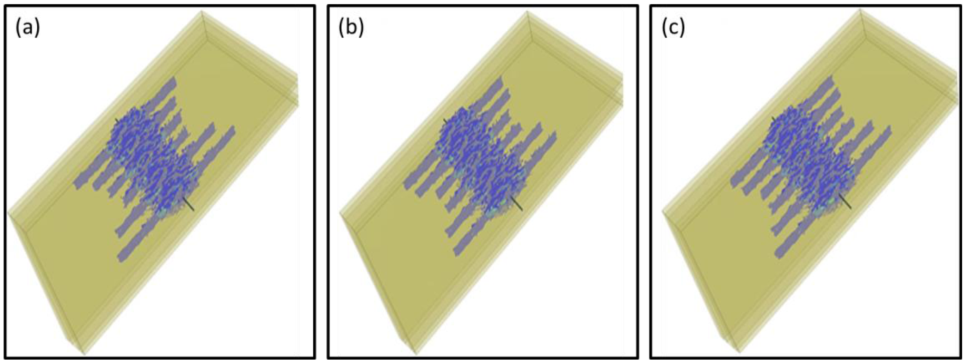

Figure 9—Fracture morphology for 6 clusters with different plugging times: (a) Plugging once; (b) Plugging twice; (c) Plugging three times.

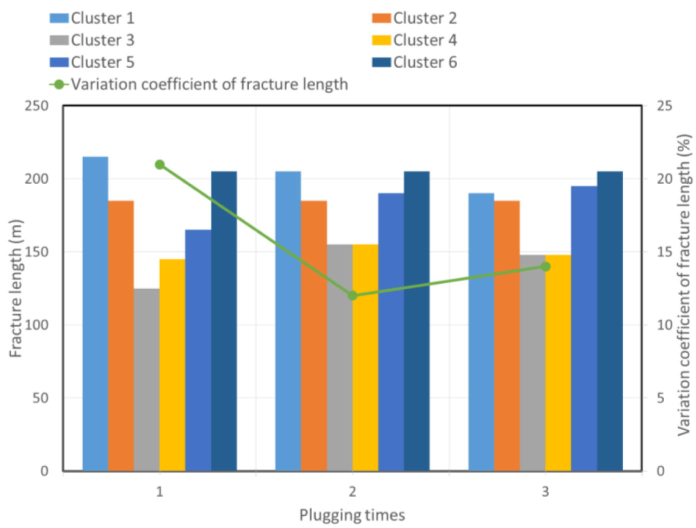

Figure 10—The variation coefficient of fracture length for 6 clusters with different plugging times: (a) Plugging once; (b) Plugging twice; (c) Plugging three times.

Fig. 11 and Fig. 12 show the fracture morphologies and calculated variation coefficient of fracture length for 7 clusters under the condition of different plugging times, respectively. The simulation results obtained are very similar to the 6 cluster conditions. The results indicate that when plugging twice, the multiple fracture propagation is the best, and the variation coefficient of fracture length is relatively lower at 16%.

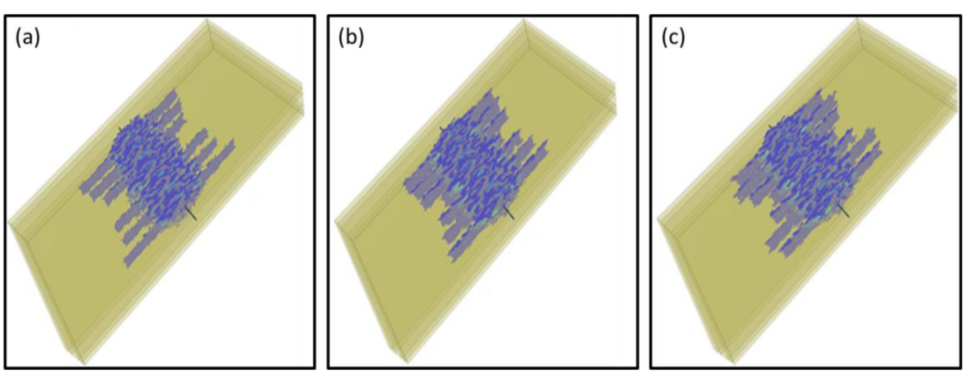

Figure 11—Fracture morphology for 7 clusters with different plugging times: (a) Plugging once; (b) Plugging twice; (c) Plugging three times.

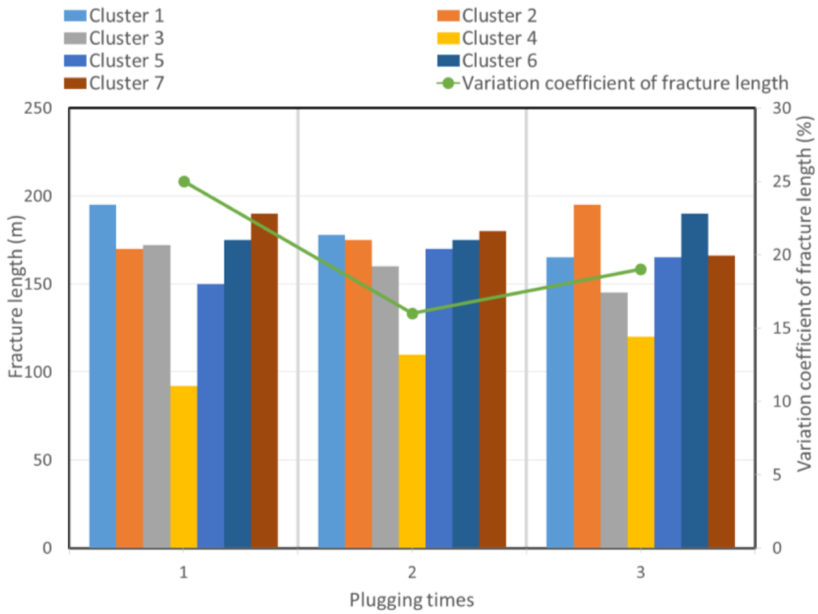

Figure 12—The variation coefficient of fracture length for 7 clusters with different plugging times: (a) Plugging once; (b) Plugging twice; (c) Plugging three times.

Amount of plugging balls. The number of plugging balls used will significantly affect the morphology of fracture propagation. However, the number of plugging ball is primarily based on empirical experience before. Fig.13 and Fig.14 show the fracture morphologies and the calculated variation coefficient of fracture length for five clusters under different number of plugging ball, respectively. The results indicate that when no plugging balls is used (Fig. 13a), the fracture lengths of the three inner fractures are significantly shorter than those of the two outer fractures, with a variation coefficient of fracture length at 39%. When 8 or 12 plugging balls within a stage is used (Fig. 13b and Fig. 13c), the outer two fractures are suppressed to a certain extent, but the fracture lengths of the inner fractures show limited improvement, and the propagation of multiple fractures remains uneven. In these cases, only a limited number of perforations are effectively plugged. As the number of plugging ball increases to 16 and 20 (Fig. 13d and Fig. 13e), the fracture morphologies under these two conditions become relatively similar, with variation coefficients of fracture length approximately 10% for both, indicating a significant improvement in the fracture uniformity. In these two cases, most of the preferred-entry perforation clusters have been effectively blocked. However, when the number of plugging ball is further increased to 24 (Fig. 13f), the lengths of the two inner fractures grow abruptly, even exceeding those of the outer fractures, resulting in a variation coefficient of fracture length at 34%, and the multiple fracture propagation becomes uneven again. Overall, with the increase of number of plugging balls, the variation coefficient of fracture length first decreases and then increases, and 16-20 plugging balls (53%-67% of total perforation hole number) are the optimal usage.

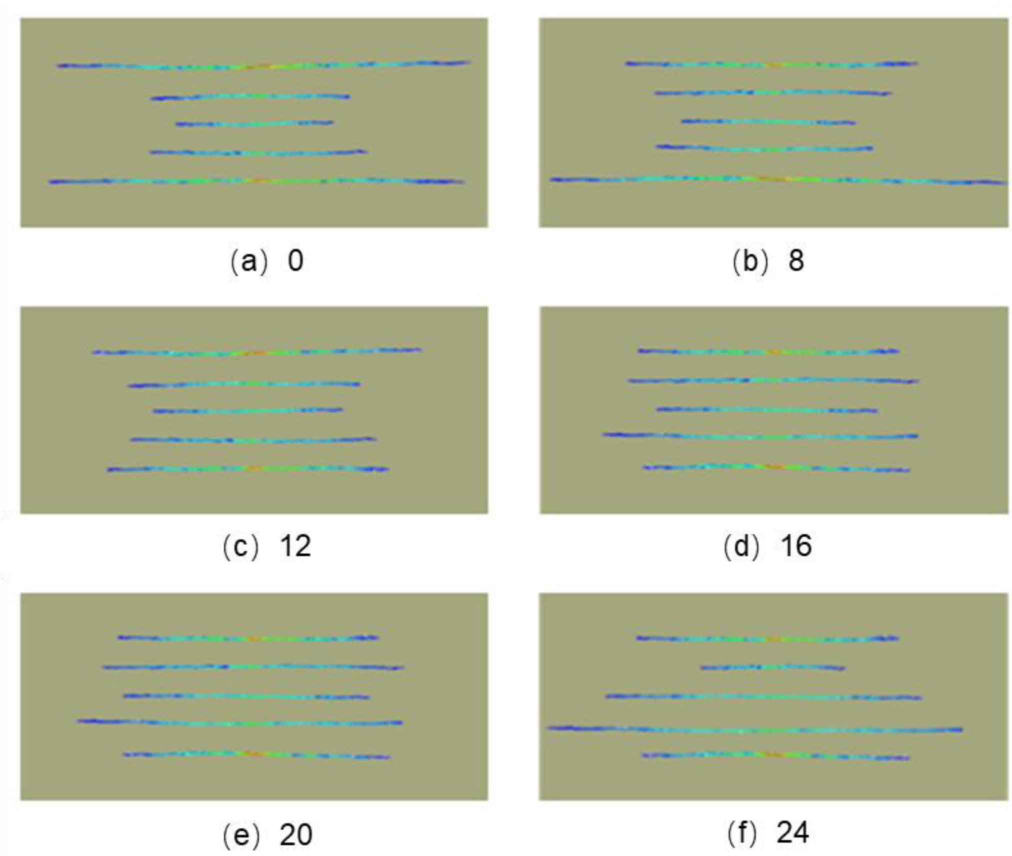

Figure 13—Fracture morphology under different number of plugging ball: (a) 0; (b) 8; (c) 12; (d) 16; (e) 20; (f) 24.

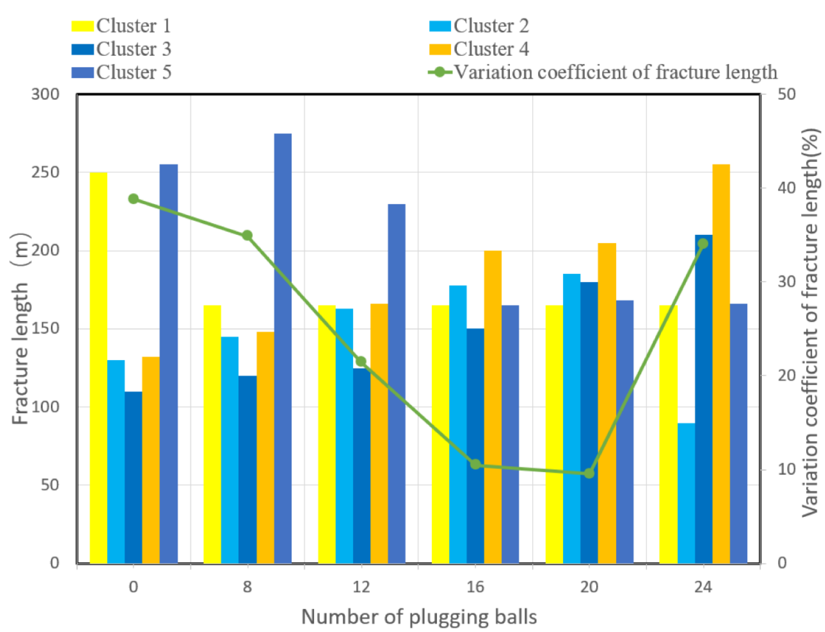

Figure 14—The variation coefficient of fracture length under different number of plugging ball: (a) 0; (b) 8; (c) 12; (d) 16; (e) 20; (f) 24.

Plugging timing. Current field practices for temporary plugging primarily rely on empirical approaches, generally pumping plugging balls when 50% of the fracturing fluid has been injected. This study presents a systematic investigation of optimal plugging timing. Fig. 15 and Fig. 16 illustrate the fracture morphologies and calculated variation coefficient of fracture length for five clusters under different plugging timing, respectively. The results indicate that when plugging balls are injected after 30-40% of the total fluid volume (Fig. 15b and Fig. 15c), compared to the non-plugging scenario (Fig. 15a), the outer fractures are significantly suppressed while the inner three fractures exhibit obvious greater lengths. These three scenarios all show uneven fracture geometry, with variation coefficients of fracture length exceeding 35%. In contrast, when plugging balls are injected at 60-70% of the total fluid volume (Fig. 15e and Fig. 15f), the variation coefficient of fracture length is relatively smaller (below 10%), indicating significantly improved fracture uniformity. The above research shows that if the plugging balls are pumped too early, much more fracturing fluid will flow into the inner fractures. In this case, the length of inner fractures will be greater than that of outer fractures, resulting in the increase of variation coefficient. It can be seen that pumping temporary plugging balls at the 50% of total fluid amount is not the optimal choice. Comprehensively, it is recommended to pump the plugging balls when 60-70% of fracturing fluid is injected. And for plugging twice, the simulation results show that the balls should be pumped at the fluid usage of 40% and 70%, respectively.

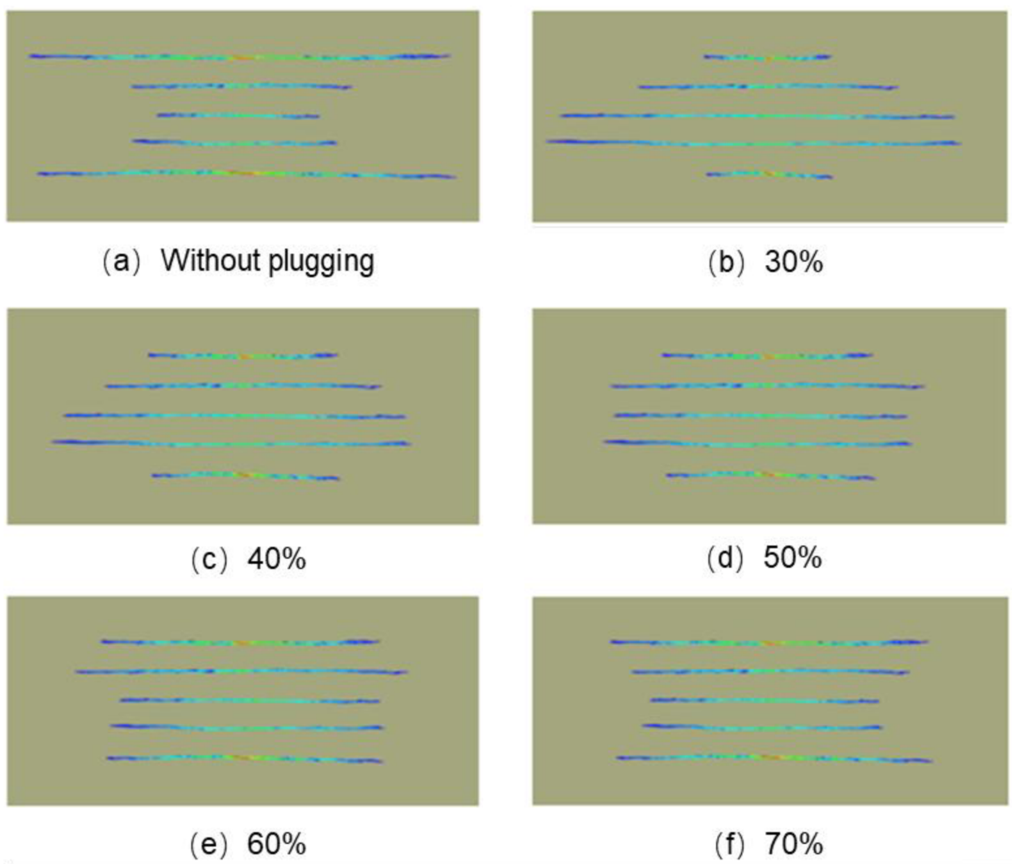

Figure 15—Fracture morphology under different plugging timing: (a) Without plugging; (b) 30%; (c) 40%; (d) 50%; (e) 60%; (f) 70%.

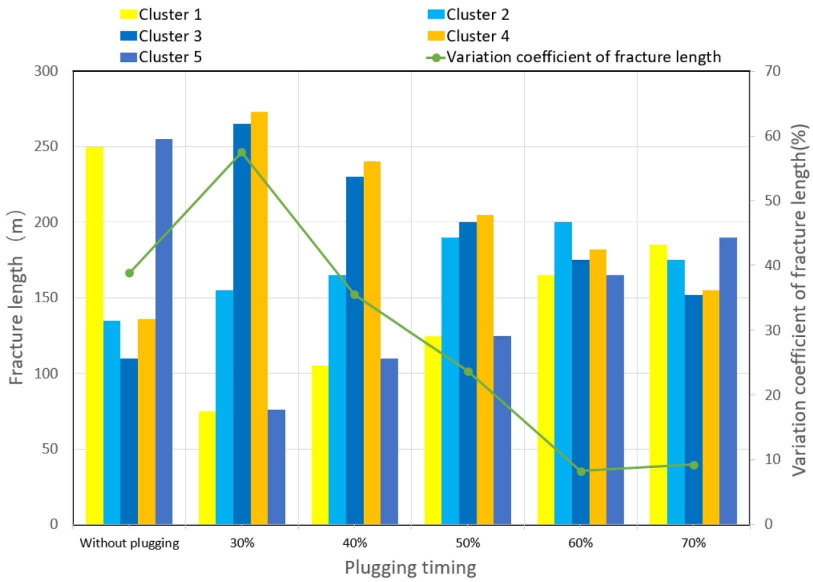

Figure 16—The variation coefficient of fracture length under different plugging timing: (a) Without plugging; (b) 30%; (c) 40%; (d) 50%; (e) 60%; (f) 70%.

Case study

The Jurassic SXM Formation in the central Sichuan Basin, China, develops multiple sets of channel sand bodies, which host abundant tight oil resources. The SXM Formation reservoir is a typical normally pressured reservoir, primarily composed of medium-fine to coarse-grained sandstones with moderate sorting and predominantly grain-to-grain line contacts. The reservoir space is dominated by intergranular mixed pores, with visible oil immersion to oil spot. Laboratory measurements indicate an average porosity of 7.6% and an average permeability of 0.229mD, with an average oil saturation of 56%. The quartz content of the reservoir ranges mainly between 30% and 60%, while the clay mineral content varies from 10% to 30%. The lithology is mainly lithic feldspar sandstone, exhibiting favorable brittleness for hydraulic fracture propagation. The Young's modulus is moderate, averaging 32.3GPa, and the Poisson's ratio averages 0.22, which facilitates the generation of hydraulic fractures with higher width. The maximum horizontal principal stress is 52.8MPa, the minimum horizontal principal stress is 43.4MPa, and the vertical stress is 47.6MPa. The geo-stress state indicates a strike-slip stress regime in this region.

Well X1, a horizontal well deployed in the SXM Formation in central Sichuan Basin with a vertical depth of approximately 2100m, was developed using multi-cluster fracturing technology. Due to the underdeveloped natural fractures in the reservoir, a complex fracture network can not be established through hydraulic fracturing. Instead, the well primarily relies on generating multiple long hydraulic fractures in each stage to reduce the flow distance of crude oil. The horizontal section of Well X1 has a total length of 1200m, with an average stage length of 70 m. Each stage consists of 5-8 clusters, and the injection rate was maintained at 20m^3/min. The fracturing fluid employed was slick-water, and the used proppant consisted of a combination of 40/70-mesh and 30/50-mesh quartz sand. To enhance stimulation effectiveness, temporary plugging fracturing was applied. Based on the findings of this study, to improve multiple fracture propagation, the design for Well X1 involved deploying temporary plugging balls in each stage at approximately 60% of the perforation number, generally plugging once in a stage. The plugging balls were injected from the wellhead when about 60% of fracturing fluid is injected.

The well successfully completed 18 stages of fracturing operations with stable treatment pressure curves, maintaining a pumping pressure of 48-57MPa. Based on the pressure variations before and after temporary plugging, the plugging efficiency was determined to be 78%, indicating that the injected plugging balls effectively formed bridging seals at the perforations, achieving the fluid diversion. Microseismic monitoring revealed that the average stimulated reservoir volume (SRV) per stage was $81\times10^4 m^3$, with an average hydraulic fracture length of 400m. The microseismic events effective cover the channel sand bodies, and the variation coefficient of fracture length per stage was controlled below 10%, ensuring uniform multiple fracture propagation. After fracturing treatment, the well underwent an extended shut-in period to facilitate oil imbibition and displacement. Subsequent production testing yielded a high industrial oil flow, marking a major breakthrough in tight oil development in this region.

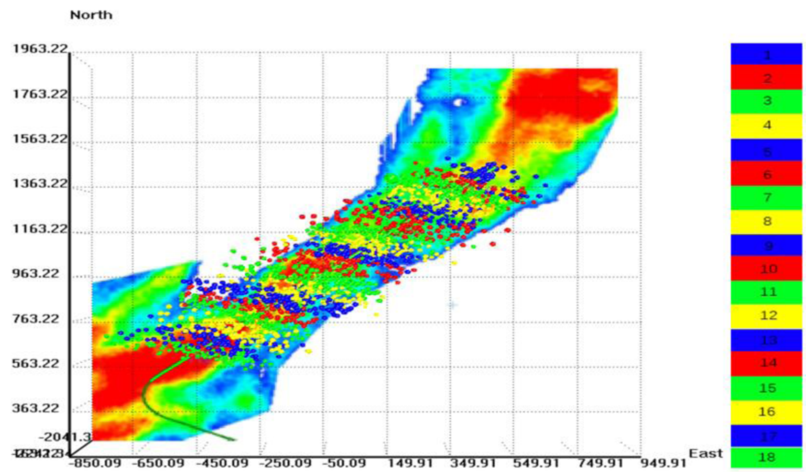

Figure 17—Microseismic monitoring results of Well X1

Conclusion

In this paper, a 3D discrete lattice modeling approach is used to simulate multi-cluster hydraulic fracturing in horizontal wells. Based on the model, the effects of perforation parameters and temporary plugging parameters on multiple fracture propagation within a stage are studied. The variation coefficient of fracture length is applied to quantitatively evaluate the fracture uniformity. The main conclusions are as follows:

- The simulation results demonstrate that reducing both the number and diameter of perforations can effectively enhance perforation friction, thereby promoting even fluid distribution among multiple clusters through limited-entry effects and improving fracture uniformity. Analysis suggests an optimal configuration of 6 perforations per cluster with a maximum diameter of 8mm for balancing multiple fracture propagation.
- When there are 5 clusters within a stage, only one plugging operation is needed, and for 6 or 7 clusters, plugging twice is necessary to ensure the even fracture propagation. With the increase of number of plugging ball, the variation coefficient of fracture length first decreases and then increases, and 16-20 plugging balls (53%-67% of total perforation hole number) are the optimal usage. It is recommended to pump the plugging balls when 60-70% of fracturing fluid is injected.
- The research findings of this study have guided the design of temporary plugging parameters for tight oil horizontal well fracturing in the central Sichuan Basin, China. Field test data demonstrate successful plugging efficiency, with a variation coefficient of fracture length below 10%, indicating effective reservoir stimulation. The well achieved high industrial oil production, conclusively validating the applicability of the proposed methodology.

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